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Subsea Inspection Supported by AI-Driven Computer Vision

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Abstract

The integrity management of subsea pipelines requires manual, time-consuming, and subjective review of vast quantities of underwater imagery, creating significant operational bottlenecks and costs for asset operators. This report presents the successful development and validation of a Human-in-the-Loop (HITL) AI system designed to augment expert inspectors by automating the initial screening of this data. The core objective was to create a robust deep learning model to accurately triage images, deprioritising those depicting healthy assets and flagging potential anomalies for focused human review. The methodology centered on a transfer learning approach, fine- tuning a MobileNetV2 architecture on an expert-annotated dataset of 2,166 real-world inspection images. A key technical challenge was a severe class imbalance, with anomalies requiring review constituting only about 8% of the data. This was systematically addressed through class weighting, a two-phase training strategy, and the implementation of an EarlyStopping callback to prevent model overfitting and ensure robust generalisation.

The final, optimised AI model demonstrated a significant success, proving its value as a powerful tool for inspectors by delivering substantial efficiency gains. The system's primary impact is its ability to drastically reduce the manual review burden. For high- consequence assets, a "Maximum Safety" mode successfully identified 100% of all true anomalies in the test set, guaranteeing no potential threat is missed. Even in this most cautious setting (Maximum Safety mode), the system still delivered a significant 30% reduction in manual labour, proving that safety and efficiency gains are not mutually exclusive. In a balanced configuration, the model reduces the required human workload by a 75%, while still correctly identifying over two-thirds (68%) of all true anomalies. Critically, the system's sensitivity can be precisely tuned to meet specific operational needs and risk levels. This proven ability to configure the AI's behaviour is a core success of the project. The successful outcome of this proof-of-concept establishes a clear, data-driven pathway to reducing inspection cycle times from weeks to hours, improving the consistency of assessments, reduce the risk of sustained attention fatigue leading to critical anomalies being missed, and enabling a proactive approach to pipeline integrity management, all while preserving essential human judgment for all final decisions.

Introduction

Ensuring the structural integrity and operational safety of subsea pipeline networks remains a concern for the global energy sector. These critical assets are exposed to harsh conditions, leading to degradation mechanisms like corrosion, cracking, and mechanical damage, which pose significant financial, environmental, and safety risks. For decades, inspection protocols have relied on established, yet often time-consuming, manual visual survey methods.

The core challenges of underwater inspection remain. Light attenuation and colour distortion in aquatic environments, backscatter from particulate matter, and the obscuring effects of biofouling present persistent obstacles to clear visual data acquisition (Bastian n.d.) (Cha n.d.). Conventional inspection campaigns, which generate vast quantities of video data, have traditionally required extensive human review, a process that is not only labour-intensive but can also delay the identification of critical defects by months (Dung 2019). There is also the risk that human inspectors may miss observing and reporting critical anomalies due to sustained attention fatigue, while they spend many hours (in some cases, hundreds of hours) observing similar looking video footage (Skinner and Giesbrecht 2025).

The response to this challenge utilises intelligent robotic platforms with AI-driven analytics to support, rather than supplant, human expertise. While advances in computer vision and real-time analysis have brought us closer to autonomous inspection, the complexities of subsea environments and the high stakes of integrity assessments mean that human oversight remains essential. Recent trends favour human-in-the-loop systems, in which AI handles initial triage and prioritisation, enabling expert inspectors to focus on the most critical or ambiguous cases.

Modern systems now leverage state-of-the-art object detectors like YOLOv5 and segmentation models such as Mask R-CNN, which are specifically fine-tuned to identify and precisely delineate a wide array of defect types under challenging visual conditions. (Cha n.d.) This progress is accelerated by advanced machine learning strategies, including the use of Generative Adversarial Networks (GANs) to create synthetic training data and domain adaptation techniques that allow models to generalise across different geographical and environmental conditions.

Perhaps the most significant recent shift is the move from cloud-based post-processing to real-time, on-board analysis. The deployment of powerful edge computing platforms, such as the NVIDIA Jetson series, on Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs) enables immediate, in-situ defect detection (Hoang ND 2019). This allows for real-time alerts to operators and dynamic mission adjustments, transforming the inspection process from a retrospective analysis to an interactive, data-driven operation. Furthermore, the fusion of optical data with other sensing modalities, particularly sonar and laser scanning, is creating comprehensive 3D digital models of pipeline segments (Vedhus Hoskere 2018).

This integration of real-time data from multiple sources allows the creation of "digital twin" concepts for asset integrity management (Ortiz 2016). A digital twin is a dynamic virtual replica of a physical asset, continuously updated with operational and inspection data. This allows for not just the detection of existing flaws, but the prediction of future vulnerabilities, the optimisation of maintenance schedules, and the simulation of asset performance under various scenarios (Samnejad 2020).

This report examines the landscape of AI-augmented subsea inspection, focusing on how deep learning can reduce manual workload, improve response times, and enhance consistency —while maintaining human involvement in final decision-making.

The objectives of this report are to:

- **Establish an objective threat classification framework** for subsea pipeline anomalies, collaboratively defined with Subject Matter Experts and tailored for use with subsea imagery datasets.

- **Develop and implement an AI-based image classification pipeline** by leveraging transfer learning and state-of-the-art deep learning architectures for AI assisted automated threat assessment.
- **Optimise the AI model through systematic training and fine-tuning,** employing robust data augmentation, stratified sampling, and best practices in model checkpointing to enhance performance and generalisability.
- **Comprehensively evaluate model performance** using quantitative (accuracy, confusion matrix, precision, recall, F1-score) and qualitative (error analysis, interpretability metrics to assess reliability and trustworthiness for operational deployment.
- **Assess the practical implications and industry relevance** of the developed system in reducing manual inspection effort, improving response times, and laying groundwork for collaborative, digitally enabled pipeline management solutions.

Terminology

- *Accuracy:* The overall percentage of correct predictions the model made across all images.
- *Decision Threshold:* The adjustable "confidence score" (from 0.0 to 1.0) the AI needs to reach before it flags an image for review. This is the main control for tuning the AI's sensitivity.
- *Recall:* The percentage of actual anomalies that the model successfully found. A high Recall minimizes the risk of missing a real threat.
- *Precision:* The percentage of images flagged by the AI that were correct. A high Precision reduces the number of false alarms an inspector has to review.
- *False Positive (FP):* The AI falsely flags the anomaly image as a potential threat.
- *False Negative (FN):* The AI dismisses the anomaly image that contains a real anomaly, marking it as "No Threat."
- *Class Imbalance:* The significant inequality in the dataset, where "No Threat" images (the majority class) vastly outnumber images with anomalies (the minority class).
- *Overfitting:* An error where the model memorizes the training data instead of learning to generalize, leading to poor performance on new, unseen images.

Methodology

This study employed a structured, four-phase methodology to develop and validate a human-in-the-loop (HITL) AI classification system for assessing pipeline anomalies from subsea imagery. The system is explicitly designed to support expert inspectors by automatically classifying images, flagging only those requiring further human review, and continuously improving through inspector feedback (see Figure 1). Model development used Python with TensorFlow 2.x and the Keras API.

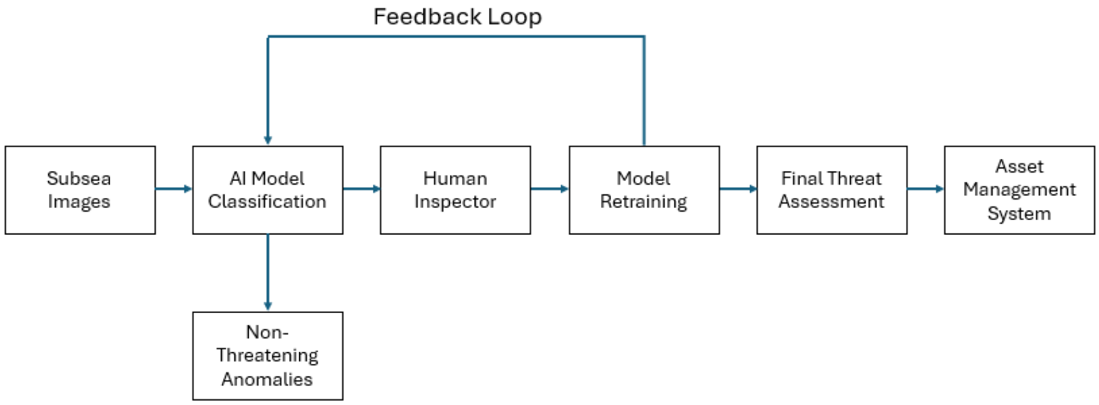

Figure 1—This figure illustrates the human-in-the-loop (HITL) workflow for AI-assisted subsea inspection. The process begins with (a) initial AI model classification of subsea images, which filters out non-threatening anomalies. (b) Images flagged by the AI are then passed to a human inspector for review. (c) The inspector's feedback is used to retrain and improve the model, closing the feedback loop and leading to a final threat assessment that is integrated into the asset management system.

Phase 1: Dataset Curation and Threat-Level Definition (Human Expertise Integration)

The initial phase involved close collaboration with subsea pipeline engineering Subject Matter Experts (SMEs) to define an actionable and objective labelling framework tailored for HITL workflows.

- **Binary Threat-Level Taxonomy:** SMEs established a two-class system:
- **No Threat:** Images where detected anomalies are considered irrelevant for operational safety.
- **Flag for Human Review:** Images containing anomalies that may pose a risk or require expert assessment. This taxonomy ensures the AI's outputs are directly actionable, streamlining human review.
- **Image-Level Classification:** Each image in the dataset was reviewed and assigned a single label according to the binary taxonomy. This approach focused on distinguishing between routine cases and those warranting expert attention.
- **Data Curation and Annotation:** SMEs conducted manual annotation of images from the archive, ensuring high-quality, domain-relevant ground truth data. Labels were stored in a metadata manifest mapping each image to its assigned class.
- **Dataset Splitting:** The annotated dataset was partitioned into training (70%), validation (15%), and test (15%) sets using stratified sampling, preserving the proportion of both classes for reliable model assessment.

Phase 2: AI Model Architecture and Data Pipeline (Human-Centric Design)

The data pipeline and model architecture were optimised for binary classification and HITL integration.

Data Input and Augmentation: Images were pre-processed and augmented (random flips, rotations, colour jitter) to improve the model's robustness across diverse inspection conditions.

1. Transfer Learning: The MobileNetV2 architecture, pre-trained on the vast ImageNet dataset, was chosen as the model and were fine-tuned to distinguish between "No Threat" and "Flag for Human Review."
2. Model Head Adaptation: The final classification layer consisted of a single output neuron with a sigmoid activation, representing the probability that an image should be flagged for human review.

Phase 3: Model Training and Optimisation (Iterative Human Feedback)

The training process was designed for continuous improvement and future adaptability to inspector feedback.

1. **Loss Function and Optimiser:** Binary cross-entropy and the Adam optimiser were used for stable and effective learning.
2. **Fine-Tuning Strategy:** The classification head was initially trained with the backbone frozen, followed by end-to-end fine- tuning. Model checkpoints ensured the best validation performance.
3. **Validation and Feedback:** Performance was evaluated on the validation set, with the workflow designed to capture inspector corrections for future retraining—closing the HITL feedback loop.

Phase 4: Performance Evaluation and HITL Validation (Supporting Human Decision-Making)
Evaluation focused on both quantitative performance and qualitative interpretability to empower inspectors.

1. **Quantitative Metrics:** Binary classification metrics (accuracy, precision, recall, F1-score, confusion matrix) were used to measure the model's ability to correctly triage images for human review.
2. **Qualitative Analysis and Interpretability:** Misclassified or uncertain images were manually reviewed, and visualisations were generated to provide inspectors with insight into the model's decision-making.
3. **Operational Workflow:** In deployment, the AI automatically filters images, flagging only those requiring expert review. Inspectors assess flagged images, confirm or correct the AI's judgment, and their feedback is recorded to further refine the model.

By employing this focused, two-class HITL approach, the methodology ensures efficient, reliable triage of inspection imagery— maximising expert productivity while maintaining rigorous human oversight and continual system improvement.


Figure 2—Left: Anomaly which will be highlighted as threatening and require SME review to complete assessment. Right: Boulder Anomaly deemed non-threatening and will not require further review.

Quantifying the Impact on Human Labour

While Precision and Recall are standard technical metrics, their true value is realised when they are translated into operational impact— specifically, the quantifiable reduction in manual labour. The AI system creates efficiency gains by automatically clearing the high volume of 'No Threat' images, allowing human inspectors to focus exclusively on the smaller subset of images that the model has flagged as potentially containing an anomaly. This approach also works to minimise one of the real pipeline integrity threats from a human-centric inspection approach - that inspectors may miss a critical anomaly due to fatigue or vigilance detriment over long hours of watching similar looking footage, allowing inspectors to focus their attention on issues pre-flagged as requiring review.

Calculation Methodology

The reduction in workload is calculated based on the number of images the AI flags for review compared to the total number of images. This calculation relies directly on the Precision and Recall metrics from the test set). The two-step process is as follows:

1. **Calculate the Total Number of Flagged Images:** First, we use the Recall to determine how many of the true anomalies the model successfully found at a given threshold. Then, we use the Precision at that same threshold to determine the total size of the review queue. The formula is:

$$Total\ \ Flagged\ \ Images = \frac{(Number\ \ of\ \ True\ \ Anomalies\ \ Found)}{Precision}$$

2. **Calculate the Workload Reduction:** The percentage of labour saved is the proportion of images that were automatically cleared by the AI and did not require human review. The formula is:

$$Workload\ \ Reduction(\%) = \frac{(Total\ \ Images - Flagged\ \ Images)}{Total\ \ Images}$$

Results

The structured training methodology, particularly the inclusion of the EarlyStopping callback, proved highly effective. The fine-tuning process was automatically concluded after a total of 30 epochs, with the system restoring the model's weights from epoch 28, which represented the point of optimal generalisation before any performance degradation on the validation set occurred. The final, best-performing model was then rigorously evaluated on the unseen test set, which contained 300 ‘N’ samples and 25 ‘Y’ samples.

Performance at Default Operational Threshold

At the standard decision threshold of 0.5, the model achieved a strong baseline performance, validating the overall approach. The results are summarised below:

Table 1—Final Model Performance on the Held-Out Test Set at the Default 0.5 Decision Threshold. The metrics demonstrate the model's strong capability in identifying over half of all true anomalies (Recall), validating its utility as a screening tool despite the expected lower Precision resulting from severe class imbalance.

Metric	Score	Interpretation in Context
Accuracy	84%	The model provided the correct classification for over 83% of the images. While high, this metric is less informative than Precision and Recall due to the class imbalance.
Precision	24%	When the model raised an alert ("Flag for Human Review"), it was correct approximately one out of every four times. The remaining three were false positives.
Recall	52%	The model successfully detected and flagged **52%—more than half—**of all the genuine anomalies present in the test set that required expert review.

The most significant result here is the Recall of 52.00%. In the context of a highly imbalanced dataset where a naive model could achieve 92% accuracy by simply guessing the majority class (and thus having 0% Recall), this outcome is a clear success. It proves that the combination of class weighting and fine-tuning effectively taught the model to recognise the complex and varied features of the rare ‘Y’ class. This performance level confirms the system's viability as an initial screening tool, capable of significantly reducing the manual review burden by automatically clearing a large portion of non-critical imagery.

Threshold Analysis: A Control Panel for Operational Performance

The true value of this AI system lies in its operational flexibility, which is revealed by the threshold analysis. By adjusting the probability threshold required to flag an image, the system's behaviour can be precisely tuned to meet different business and safety objectives. The table below presents the model's performance on the test set across a range of potential thresholds.

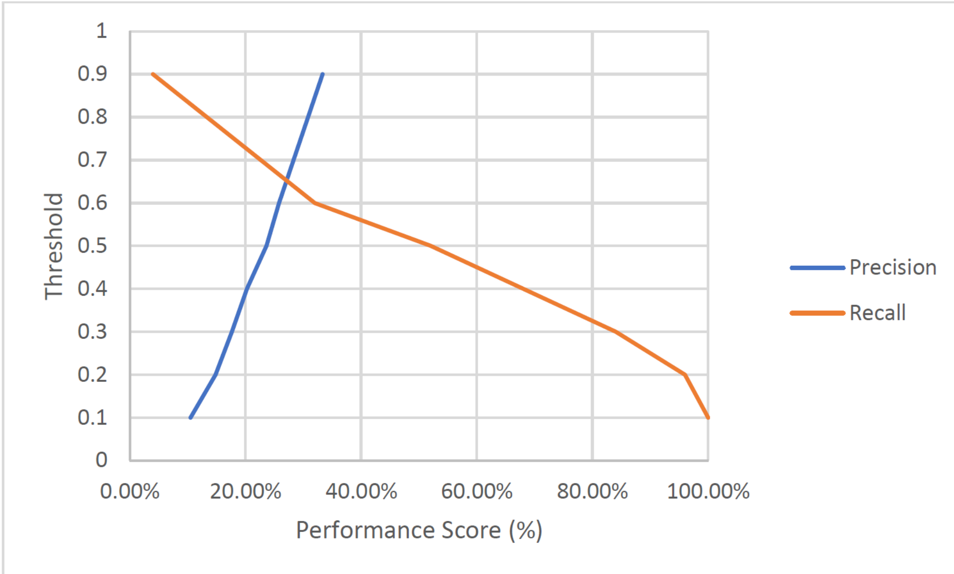

Figure 3—This plot visualises the direct trade-off between the model's two key performance metrics on the test set. The Recall curve (blue) represents the safety metric, showing the percentage of true anomalies found. The Precision curve (orange) represents the efficiency metric, showing the percentage of correct alerts. By selecting a threshold, an operator can choose their desired balance: a low threshold for a 'Maximum Safety' mode that finds all anomalies (high Recall) at the cost of more false alarms (low Precision), or a higher threshold for a 'High-Confidence' mode that minimises false alarms.

Table 2—Operational Performance Analysis at Various Decision Thresholds on the Test Set. This table illustrates the critical trade-off between Precision (reducing false alarms) and Recall (finding all true anomalies), effectively serving as a 'control panel' for tuning the AI's behavior. The results demonstrate that the system can be configured to meet specific operational needs, from a maximum safety setting (100% Recall) to a high confidence alerting mode that minimises false positives.

DecisionThreshold	Precision	Recall	WorkloadReduction	Business Use Case & Interpretation
0.1	11%	100%	30%	**Maximum Safety / High-Sensitivity Screening:** At this level, the model is configured to be extremely cautious. It successfully finds every single true anomaly in the test set. The trade-off is a very high rate of false alarms, but for the most critical assets, where missing any anomaly is unacceptable, this provides an unparalleled safety net.
0.2	15%	96%	51%	
0.3	18%	84%	64%	
0.4	21%	68%	75%	**Balanced Performance / Standard Workflow:** This threshold represents an excellent compromise. The model still catches over two-thirds of the true anomalies (68.00% Recall) while improving the Precision to over 20%. This significantly reduces the inspector's workload while maintaining a high detection rate for common issues.
0.5	24%	52.00%	83%	The default baseline, catching over half of the anomalies.
0.6	26%	32.00%	90%	**High-Confidence Alerting:** This setting maximises precision. While it will only find about a third of the anomalies, the alerts it does generate have the highest probability of being correct. This is useful when inspector time is extremely limited, and the goal is to only review the most certain and obvious threats.
0.9	34%	4.00%	99%	

Sample:

For a case where contained 325 total images, 25 of which were true anomalies with 100% recall and 11% precision.

$$Total\ \ Flagged\ \ Images = \frac{100\%*25}{11\%} = 227 images\ \ flagged\ \ to\ \ have\ \ anomalies$$

$$Workload\ \ Reduction(\%) = \frac{325-227}{325} = 30\%$$

This analysis is the key deliverable for operational deployment. It transforms the model from a black box with a single performance score into a transparent and tunable tool. Stakeholders can now have an informed discussion to select the threshold that best aligns with their operational priorities, whether that is maximising safety, optimising inspector efficiency, or a balanced approach between the two.

For instance, a company could apply a 0.2 threshold for its most critical, high-risk pipelines and a 0.5 threshold for its lower-risk assets, all using the same underlying AI model.

A Scalable Approach to Anomaly Detection

highly granular object detection models, as illustrated in, draw precise bounding boxes and apply specific labels to individual anomalies within an image. While technically impressive, this object detection approach was found to be extremely labour-intensive to scale. The process of manually drawing precise bounding boxes and masks for thousands of images to create a suitable training dataset proved to be a significant bottleneck, making it difficult to develop a robust, generalisable model in a cost-effective and timely manner.

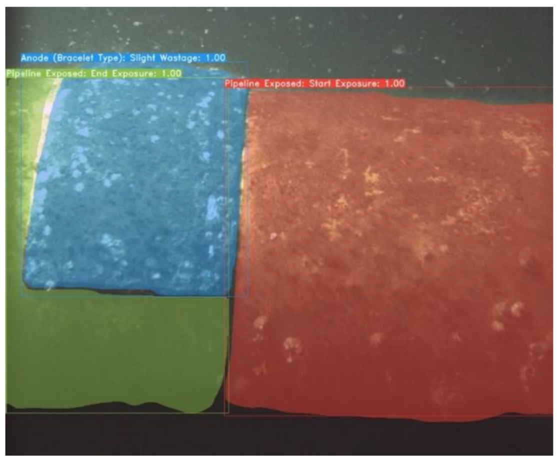

Figure 4—An example of the AI model's qualitative performance. The system correctly identifies multiple distinct features, including (a) slight wastage on a bracelet-type anode, and both the (b) start and (c) end of an exposed pipeline section. This demonstrates the model's ability to handle complex scenes.

The success of the current model validates this strategic shift. By leveraging the existing database with its simpler annotations, we have successfully trained a classifier that correctly handles visually complex scenes like the one in . Although the model does not draw the bounding boxes, its high Recall score proves that it has learned to recognise the collective features within such images and correctly assign them to the "Flag for Human Review" class.

This approach represents a significant breakthrough. It demonstrates that a scalable, efficient data labelling strategy is sufficient to build a powerful AI screening tool. The system effectively automates the most time-consuming part of the inspection process—reviewing thousands of benign images—without requiring the prohibitively expensive and time-consuming annotation process of a full object detection model. This makes the entire AI-assisted workflow viable for wide-scale deployment.

Crucially, this binary classification system serves as the perfect foundation for future work. The model's ability to flag relevant images creates a highly targeted dataset. As inspectors review these flagged images, their detailed feedback and annotations can be systematically captured. This creates a powerful, self-reinforcing loop where the output of the efficient classifier is used to build a high- quality, focused dataset for developing the more advanced object detection models in the future, solving the original scalability problem.

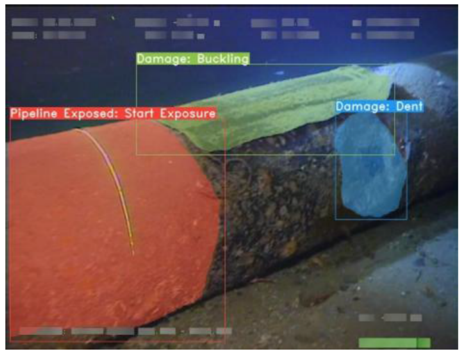

Figure 5—A demonstration of the model correctly flagging several distinct types of structural anomalies within the "Flag for Human Review" class, including (a) buckling, (b) denting, and (c) the start of pipeline exposure.

Discussion

The results of this study provide evidence for the value of AI-assisted systems in the field of subsea asset integrity management. The successful development of a tunable classification model moves the concept from a theoretical possibility to a practical tool with clear operational benefits.

Analysis of Model Performance and Error Modes

The final model's performance should be viewed through the lens of the classic Precision-Recall trade-off. In a safety-critical domain like pipeline inspection, Recall is often the more important metric. The consequence of a false negative (missing a true threat) can be catastrophic, leading to environmental damage, significant financial loss, and reputational harm. The fact that the model, at a balanced threshold, can achieve a Recall of over 50% and can be tuned to achieve 100%, is a significant achievement and the primary indicator of the project's success.

The lower Precision score (24% at default) is a typical and expected characteristic of models trained on highly imbalanced datasets. It indicates that the system will generate a number of false positives. However, in a HITL workflow, this is an acceptable, and even desirable, outcome. The model's role is not to be perfect, but to be a workload reduction tool. By filtering out the ~80-90% of images that are confidently classified as 'No Threat', the system drastically reduces the volume of data an inspector needs to review, even if some of the flagged images are ultimately benign. This allows expert human attention to be concentrated where it is most needed: on the ambiguous and potentially high-consequence cases.

A qualitative review of the model's errors (both false positives and false negatives) would be a critical next step, but we can hypothesise the likely causes:

- **False Positives (Predicted 'Y', Actually 'N'):** These errors are likely caused by visual ambiguity. For example, unusual marine growth, shadows from the ROV, or reflections could create patterns that mimic the features of genuine anomalies like corrosion or coating damage. These "hard negative" examples are invaluable for future retraining.
- **False Negatives (Predicted 'N', Actually 'Y'):** These are the more critical errors to understand. They are likely to occur with anomalies that are very subtle, small in scale, or of a rare type not well-represented in the training data. Poor image quality, such as high turbidity or motion blur, would also be a significant contributing factor.

Relevance and Impact on the Energy Industry

The adoption of a system like this offers a transformative impact on operational efficiency. The quantitative analysis shows that, when configured for a balanced performance (0.4 threshold), the AI tool reduces the required manual review workload by a remarkable 75%. Even when tuned for maximum safety (0.1 threshold), where it successfully identifies 100% of all true anomalies, the system still delivers a significant

30% reduction in labour. This ability to decouple safety assurance from manual effort is a paradigm shift for inspection workflows:

1. **Increased Operational Efficiency and Cost Reduction:** The most immediate benefit is a dramatic reduction in the man-hours required for visual inspection analysis. This accelerates the inspection-to-reporting cycle from weeks or months to days or even hours. This not only reduces direct costs associated with engineering time but also allows for a higher frequency of inspection reviews, improving overall asset oversight.
2. **More Rapid Response to Integrity Threats:** There is a growing trend to stream offshore inspection video footage to a shore- based review team to allow a more cost-effective remote inspection review team to analyse the video footage. By incorporating rapid AI real-time video classification models, anomalies can be alerted and classified rapidly for human review, allowing immediate actions by onshore engineering teams or offshore inspection vessel to be undertaken.
3. **Improved Consistency and Data Standardisation:** The AI model applies the same classification criteria to every single image, removing the inherent subjectivity and variability that exists between different human inspectors or even the same inspector over time due to fatigue. This leads to a more consistent, auditable, and standardised assessment of asset health, which is invaluable for long-term trend analysis and regulatory reporting.
4. **Enhanced Safety and Proactive Risk Mitigation:** By accelerating the identification of potential threats, the system enables faster intervention and a shift from reactive to proactive maintenance. The ability to tune the system for 100% Recall on critical assets provides an unprecedented level of safety assurance, ensuring that no potential threat goes unnoticed.
5. **Enabling Predictive Maintenance and Digital Twins:** The structured, classified data generated by the AI system is a high- value asset in itself. When integrated into asset management platforms and digital twins, this data can be used to track anomaly progression over time, predict future rates of degradation, and optimise maintenance schedules based on quantitative risk models rather than fixed time intervals. This data-driven approach is the future of asset integrity management.

Limitations and Future Work

While this proof-of-concept was highly successful, it is essential to acknowledge its limitations and outline a clear path for future development.

Current Limitations:

1. **Dataset Size and Variety:** The model's performance is fundamentally constrained by the data it was trained on. The current dataset, while valuable, is limited in size and may not represent all possible anomaly types or environmental conditions.
2. **Binary Classification:** The current ‘Y’/’N’ classification is effective for triage but does not provide specific information about the type or severity of the detected anomaly.
3. **Static Model:** The current model is a static snapshot. Without a defined process for retraining, its performance will not improve, and it will not adapt to new information.

Recommended Future Work:

1. **Conduct Systematic Error Analysis:** The highest priority next step is to perform a detailed qualitative review of the model's false positives and false negatives on the test set. This will provide invaluable insights into the model's specific weaknesses and guide future data collection efforts.
2. **Implement a Continuous Retraining Loop:** The HITL framework must be fully realised. A system should be developed to capture all inspector corrections and feedback. This new, human-validated

data should be periodically used to retrain and fine- tune the model, ensuring it continuously learns and adapts over time.

3. **Expand the Dataset with Targeted Collection:** Based on the error analysis, a targeted data collection effort should be initiated to acquire more examples of the specific anomaly types the model struggles with, as well as more "hard negative" examples that cause false positives.
4. **Evolve to Multi-Class Classification and Object Detection:** As the dataset grows, the project should evolve beyond binary classification. The next logical step would be a multi-class classification model (e.g., 'Corrosion', 'Denting', 'Freespan'). The ultimate goal would be an object detection model (e.g., using architectures like YOLO or Faster R-CNN) that can draw bounding boxes around each specific anomaly within an image, providing precise location and classification information.
5. **Develop Real Time Anomaly Detection):** Develop and refine tools and work processes for real-time on the fly video review and anomaly flagging.
6. **Pilot Deployment and Usability Studies:** The model should be integrated into a prototype review interface and put through a pilot program with inspection engineers. Gathering direct user feedback is the only way to ensure the tool is practical, intuitive, and genuinely enhances their workflow.

Conclusion

This report has detailed the successful development, training, and validation of a human-in-the-loop AI system designed to fundamentally modernise the process of subsea pipeline inspection. By adhering to a rigorous methodology that placed expert domain knowledge at the core of the data curation process and employed advanced deep learning techniques to address real-world data challenges, this project has successfully moved beyond a theoretical concept to a validated, practical proof-of-concept with clear and quantifiable value.

The primary achievement of this work is the demonstration that a deep learning model can effectively learn to distinguish between benign and potentially threatening anomalies within a highly imbalanced, real-world dataset. The final model's ability to achieve a Recall of 100% at its maximum safety setting confirms its utility as a powerful screening agent, capable of identifying all true threats and thereby reducing the manual review workload for inspection engineers.

However, the most profound outcome of this project is the revelation of the system's operational flexibility, as illustrated by the threshold analysis. The ability to tune the model's sensitivity provides a strategic "control panel," allowing the business to dynamically balance the imperative of safety (maximising Recall) against the need for efficiency (maximising Precision). This transforms the AI from a static classification tool into a dynamic asset that can be configured to meet the specific risk profiles of different assets or the resource constraints of different operational phases. Whether deployed in a "find everything" mode for critical infrastructure or a "balanced assistant" mode for routine surveys, the system provides consistent, data-driven support.

In conclusion, this human-AI partnership offers a practical, scalable, and powerful path toward safer, more efficient, and more consistent subsea pipeline operations. The system does not seek to replace the invaluable expertise of human inspectors but rather to augment and amplify it, automating the repetitive and time-consuming aspects of their work and allowing them to focus their critical judgment on the most complex and consequential cases. The successful results presented herein provide a strong and compelling basis for moving forward with pilot deployment and a detailed roadmap for future enhancements that will further embed this technology into the core of modern asset integrity management.

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